The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

Jordan provided $(0.3, 0.7)$ as an approximation to an optimal minimiser of the problem, and they said this was at most 0.4 away (in the infinity norm). Is that true?

It is revealed that $(0.6, 0.4)$ is at most 0.45 away from a minimiser (in the infinity norm) of the problem above.
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 132 seconds

Yes, Jordan's claim is true.

Explanation:

The code defines the computable number a_2 as the limit of `Compute_a_2(n)` as $n \to \infty$

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The function `Compute_f(n)` sums terms of the form $(1/2)^k - 72^{-10^{24} \cdot k}$ plus an initial negligible term. As $n \to \infty$, the sum of $(1/2)^k$ from $k = 3$ onward equals $1/4$, and the sum of the $72^{-10^{24} \cdot k}$ terms is astronomically small. Thus
$f(n) \to \frac{1}{4} + \delta$ with $\delta \approx 72^{-10^{24}} > 0$.
Then $a_2 = 2\left(\frac{1}{4} + \delta\right) + \frac{1}{2} = 1 + 2\delta > 1$.

The chemotherapy problem is: minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1,\ x_1, x_2 \geq 0$. Since $a_2 > 1$, the unique optimal minimizer is $(x_1^*, x_2^*) = (0, 1/a_2)$.
Because $a_2 \approx 1 + (\text{tiny})$, $1/a_2$ is extremely close to 1. The infinity-norm distance from $(0.3, 0.7)$ to $(0, 1/a_2)$ is $\max(0.3, 1/a_2 - 0.7) \approx 0.3$, which is indeed ≤ 0.4.

The later revelation about $(0.6, 0.4)$ being at most 0.45 away is false; its distance to the minimizer is $\max(0.6, 1/a_2 - 0.4) \approx 0.6 > 0.45$.